

Analog to Digital Converters

- i) Flash A/D Converter ✓
- ii) Dual slope converter ✗ ✓
- iii) Successive Approx. ADC ✗

Flash A/D Converter:

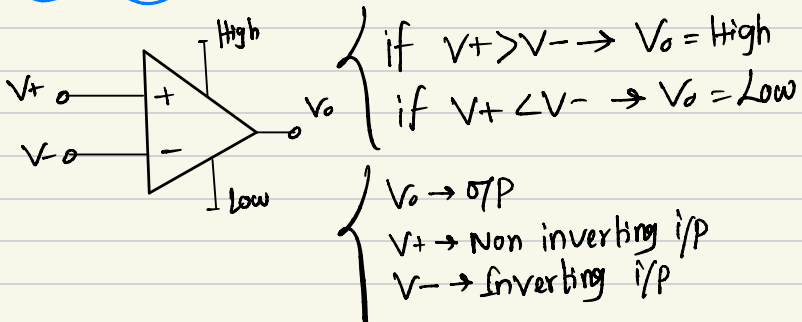
Advantages

- Fastest ADC. It requires one clock cycle to convert the analog to digital data.
- suitable for large bandwidth applications.
e.g. Satellite comm., Radar, Oscilloscope.

Disadvantages

- It requires a lot component, cost ↑
- High power consumption.
- Large Die Area
- Limited Resolution (typically up to 8 bit)

Op Amp Comparator



Let $V_{\min} = 0V$, $V_{\max} = 8V$, $n = 2$
 $\therefore$ Levels, $L = 2^n = 4$, step size, $\Delta = \frac{V_{\max} - V_{\min}}{2^n}$
 $\Delta = \frac{8 - 0}{4} = 2$

Quantization Range	Encoded/Digital signal
6-8	11
4-6	10
2-4	01
0-2	00

2-bit Flash ADC

$n = 2$, num of resistors $= 2^n = 4$

num of comparator $= 2^n - 1 = 3$

Priority encoder $2^n : n = 4 : 2$

Find V_0, V_1, V_2, V_3 using
 Voltage divider rule

$$V_0 = 0V$$

$$V_1 = \frac{R}{R+R+R+R} \times 8$$

$$= \frac{1}{4} \times 8 = 2V$$

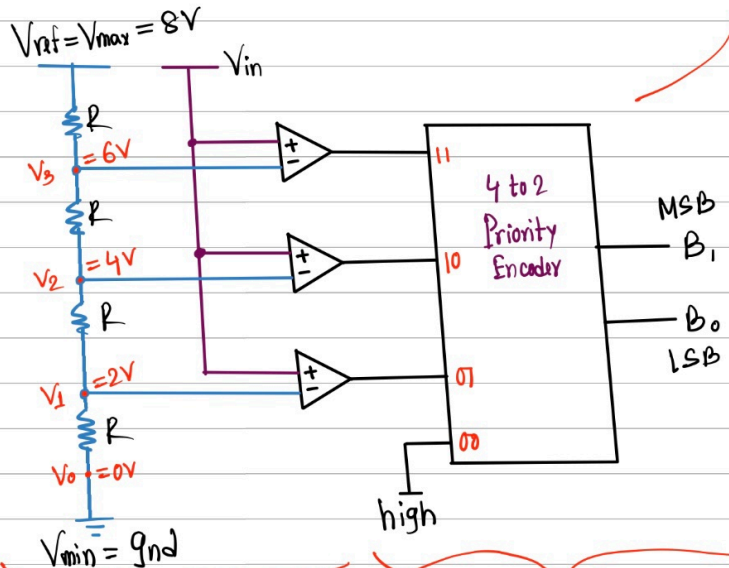
$$V_2 = \frac{R+R}{4R} \times 8$$

$$= 4V$$

$$V_3 = \frac{R+R+R}{4R} \times 8$$

$$= 6V$$

$$V_{\max} = 8V$$

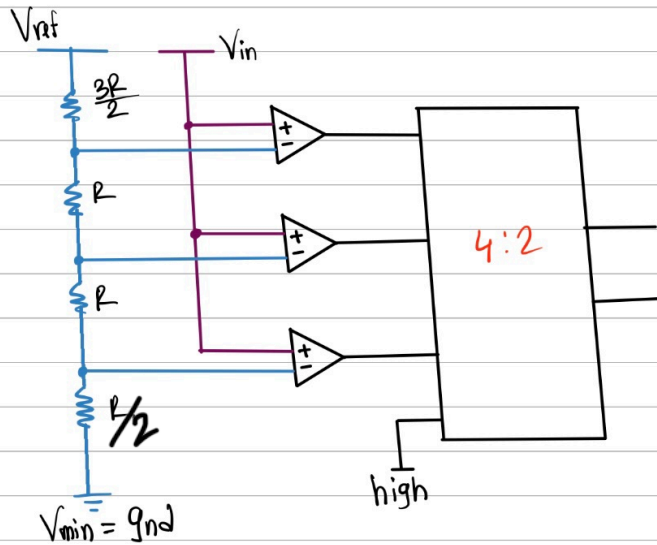


Quantization
 Range

Quantization

Encoding

Practice



For the given Flash ADC, Quantizer range is $(0.8V)$
 for the following sequences $\{0.8, 7, 3.4, 5.8, 2.2\}$
 write the encoded sequences.

Soln:

Voltage divider to find V_0, V_1, V_2, V_3

$$V_0 =$$

$$V_1 =$$

$$V_2 =$$

$$V_3 =$$

Quantization Range	i/p sequences	Encoded seq.

$\{0.8, 7, 3.4, 5.8, 2.2\}$

~~***~~ # Formulas

given $\rightarrow n, V_{\max}, V_{\min}$

$\rightarrow$ levels, $L = 2^n$

$\rightarrow$ step size / resolution, $\Delta = \frac{V_{\max} - V_{\min}}{2^n} \rightarrow 1 \text{ LSB value.}$

$\rightarrow$ Quantization error = $\frac{\Delta}{2}$
 $= \frac{1}{2} \frac{(V_{\max} - V_{\min})}{2^n}$

$\rightarrow$ Sampling period $\rightarrow T_s$

$\rightarrow$ Sampling Frequency, $F_s = \frac{1}{T_s}$

$\rightarrow$ Minimum Sampling Frequency, $F_s = 2 \times F_{\text{input}}$

$\rightarrow$ for n-bit flash ADC

$\rightarrow$ num of resistors = 2^n

$\rightarrow$ num of comparators = $2^n - 1$

$\rightarrow 2^n : n$ priority Encoder

Hardware

- 9) For an analog signal $\rightarrow V_{\min} = 0V$, $V_{\max} = 8V$ (V_{ref})
 $n = 3$ bit (i) find the quantization range, quantized value and corresponding Digital signals, (ii) find i/p op characteristics.
 (iii) Design a 3 bit Flash ADC converter

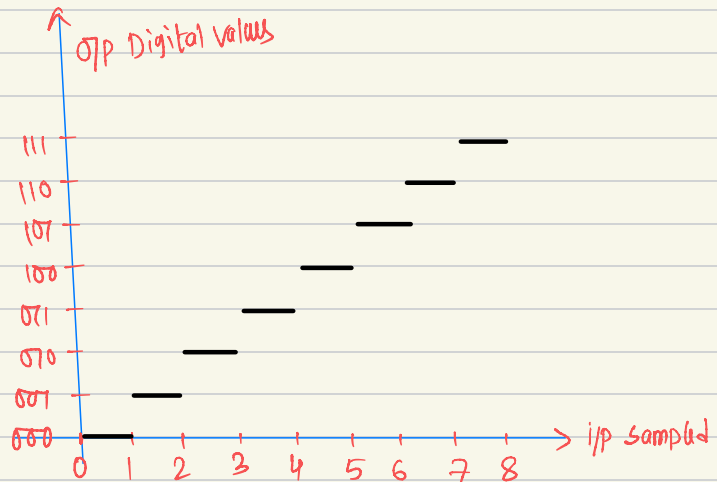
Ans:

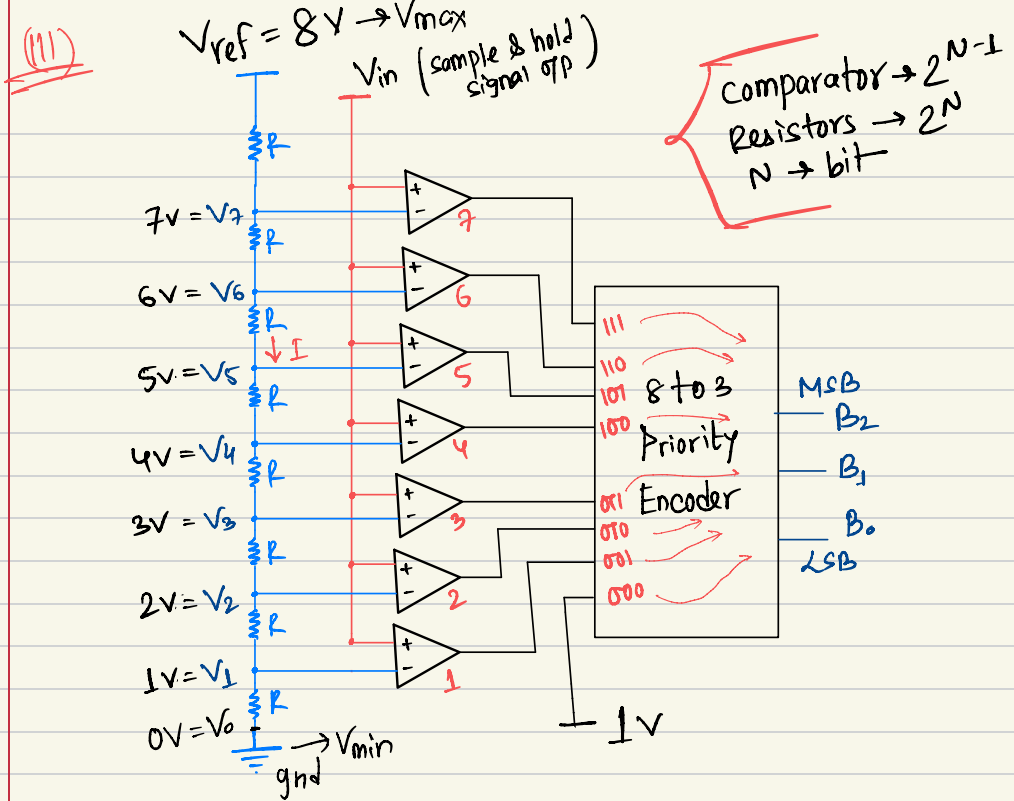
$n = 3$, $L = 2^n = 2^3 = 8$ levels. step size, $\Delta = \frac{V_{\max} - V_{\min}}{2^n} = \frac{8 - 0}{8}$
 $\Delta = 1$

(i)

Quantization Range	Quantized value	Digital/Encoded
7-8	7	111
6-7	6	110
5-6	5	101
4-5	4	100
3-4	3	011
2-3	2	010
1-2	1	001
0-1	0	000

(ii)





Quantization

Encoding

Fig: 3-bit Flash ADC

Using Voltage Divider Rule

$$V_1 = \frac{R}{R+7R} \times V_{ref}$$

$$= \frac{1}{8} \times 8 = 1V$$

$$V_2 = \frac{2R}{2R+6R} \times V_{ref}$$

$$= \frac{2}{8} \times 8 = 2V$$

$$V_3 = \frac{3R}{3R+5R} \times V_{ref}$$

$$= \frac{3}{8} \times 8 = 3V$$

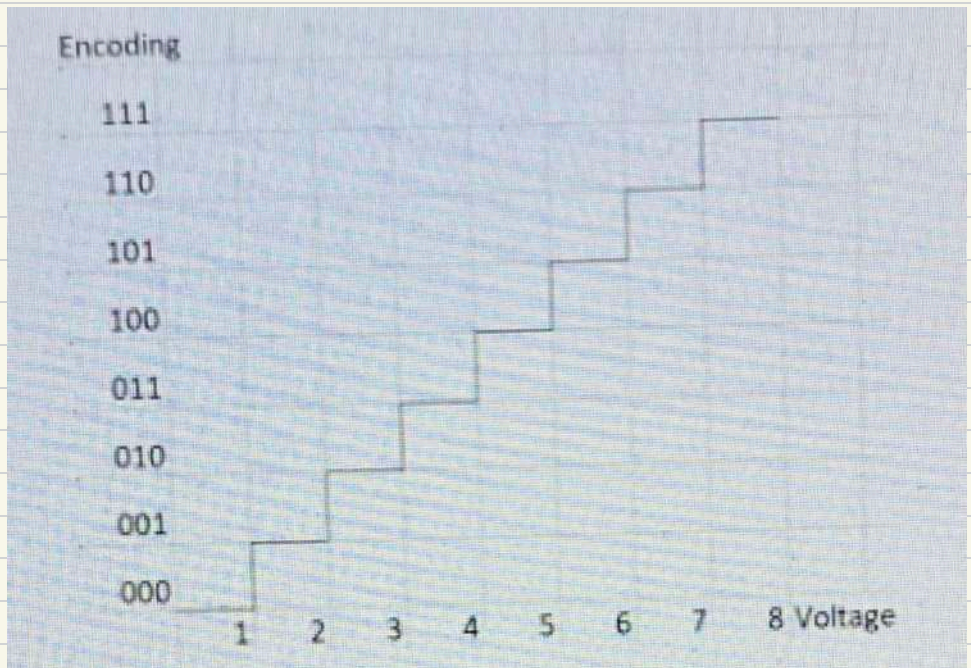
$$V_4 = \frac{4R}{4R+4R} \times 8 = 4V$$

$$V_5 = \frac{5R}{5R+3R} \times 8 = 5V$$

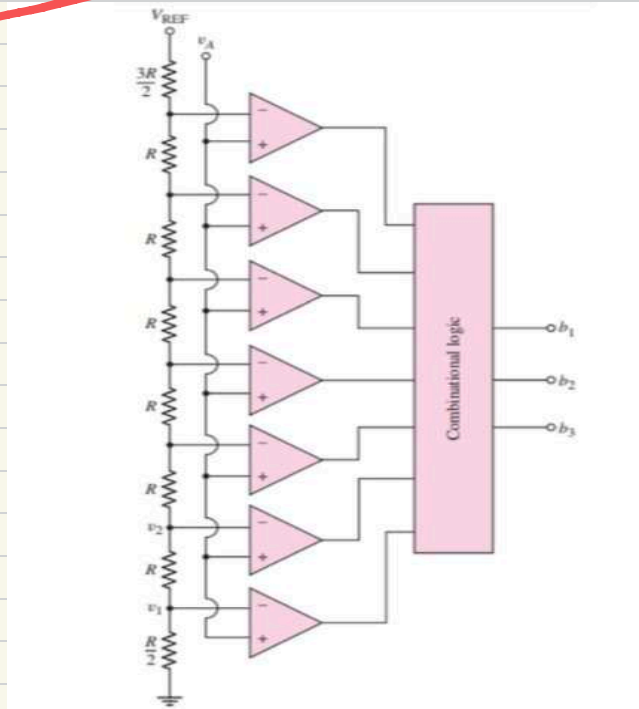
$$V_6 = \frac{6R}{6R+2R} \times 8 = 6V$$

$$V_7 = \frac{7R}{7R+R} \times 8 = 7V$$

(iii) # Input-Output Characteristics:



Practice Problem

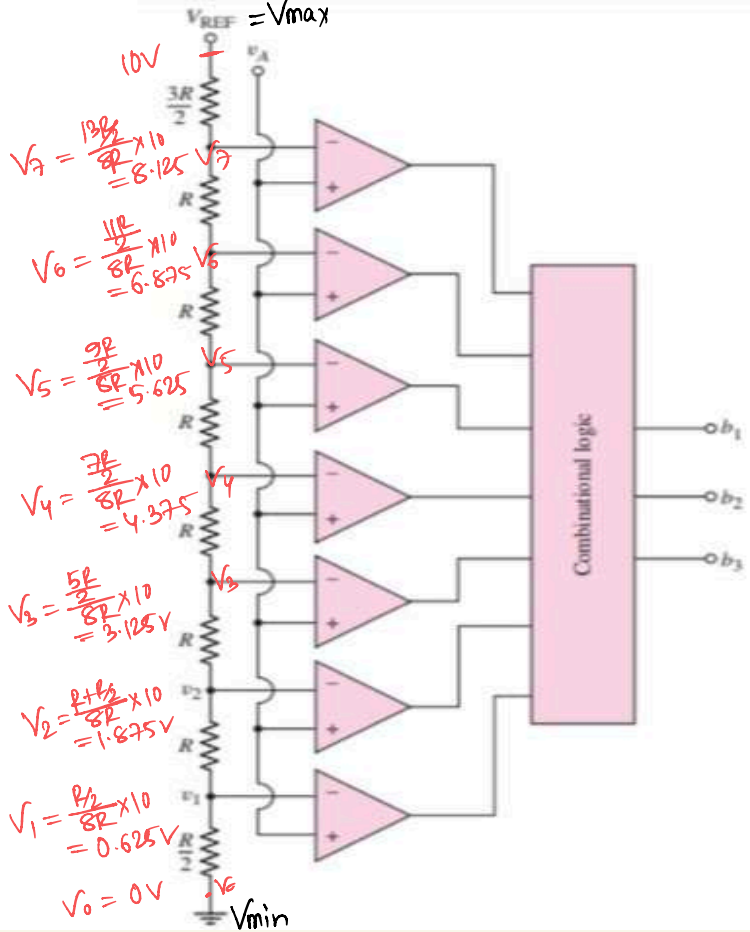


For the given Flash ADC, the reference voltage is, $V_{ref} = 10V$

- Calculate the 1 LSB value or each step size or Resolution.
- Calculate the quantization range and corresponding Digital / Encoded Signal.
- Show V_{in} vs V_{out} graph.
- Find the ranges where min and max quantization error can occur.
- If the input voltage $V_{in} = 7V$
 - Comment on the quantization range in which the i/p lies.
 - Find the digital op for the given i/p.
- If the op is '010' what is the max and min value of V_A that produces this op?
- If 1 bit was increased from this 3-bit Flash ADC, comment on the hardware changes required.

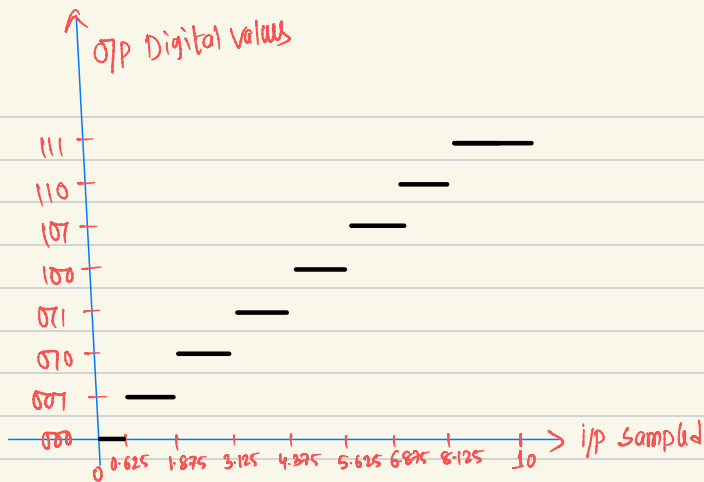
$$(a) \Delta = \frac{V_{\max} - V_{\min}}{2^n} = \frac{10 - 0}{8} = 1.25$$

(b)



Quantization Range	Encoded Signal
8.75 - 10	111
7.5 - 8.75	110
6.25 - 7.5	101
5 - 6.25	100
3.75 - 5	011
2.5 - 3.75	010
1.25 - 2.5	001
0 - 1.25	000

(c)



(d) min quantization error range $\rightarrow \boxed{0 - 0.625}$
max " " " " $\rightarrow \boxed{8.125 - 10}$
quantization error $= \frac{\Delta}{2} = \frac{1.25}{2}$

(e) (i) $\rightarrow 6.875$ to 8.125
(ii) $\rightarrow 110$

(f) min $V_A = 1.875$ V
max $V_A = 3.125$ V

(g) $n = 3 + 1 = 4$ bit

$\therefore$ number of Resistors $= 2^n = 2^4 = 16$

number of comparators $= 2^n - 1 = 2^4 - 1 = 15$

2^n : n Priority Encoder $\rightarrow 16:4$

Q. for the given Flash ADC
 $V_{ref} = 16V$.

(a) find the quantization range and corresponding digital signal.

(b) Draw the V_{ip} vs V_{op} graph.

Soln:

(a) here, $V_{max} = +V_{ref} = 16V$
 $V_{min} = -V_{ref} = -16V$, $n=4$

$$V_0 = -V_{ref} = -16V$$

$$V_j = \frac{\sum_{i=0}^j R_i}{\sum_{i=0}^n R_i} (V_{max} - V_{min}) + V_{min}$$

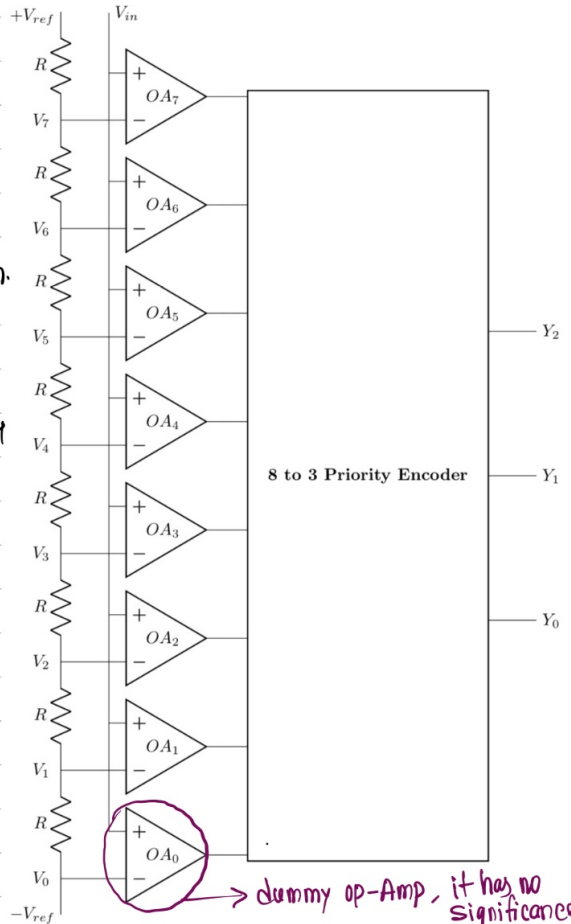
$$V_1 = \frac{R}{8R} (16 - (-16)) - 16 = -12V$$

$$V_2 = \frac{2R}{8R} (16 + 16) - 16 = -8V$$

$$V_3 = \frac{3R}{8R} (32) - 16 = -4V$$

$$\text{Similarly, } V_4 = 0V, V_5 = 4V$$

$$V_6 = 8V, V_7 = 12V$$



Quantization Range	Encoded Signal
12 to 16	111
8 to 12	110
4 to 8	101
0 to 4	100
-4 to 0	011
-8 to -4	010
-12 to -8	001
-16 to -12	000

(b)

